■# PAPER_357 — TOI-1227b: Young Neptune Exoplanet with Tidal Gravity and Disk-UQFF Coupling

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Session: *97*

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 97 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF calculation for a young exoplanet (T_age = 11 Myr) with tidal + disk force coupling **Author:** Daniel T. Murphy

<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

TOI-1227b is a rare young sub-Neptune (T_{age} = 11 Myr, P_{orb} = 11 days) still embedded in the debris disk of its M-dwarf host. UQFF computes the tidal gravitational acceleration $g_{tide} = GM_{star}/a_{orb}^2$ at the orbital radius, a disk-UQFF force $F_{disk} = \rho_{disk} \cdot v_{disk}^2 \cdot (1+H■t) \cdot SC_m$ incorporating Hubble expansion and superconductive modifier, and the full FUB_{ii}. TOI-1227b provides a benchmark for UQFF at typical planetary mass and orbital scales during the planet formation epoch.

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2. Core Physics

2.1 Tidal Gravitational Acceleration

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g_{\rm tide} = \frac{G M_\star}{a_{\rm orb}^2}
```

At a_{orb} from P_{orb} = 11 days via Kepler's third law:

```
a_{\rm orb} = \left(\frac{G M_\star P_{\rm orb}^2}{4\pi^2}\right)^{1/3}
```

For M_{star} ≈ 0.17 M■ (M-dwarf), P_{orb} = 11 d:

```
a_{\rm orb} \approx 0.05\ \mathrm{AU} = 7.5 \times 10^9\ \mathrm{m}
```

```
g_{\rm tide} \approx \frac{6.674\times 10^{-11} \times 0.17 \times 1.989\times
10^{30}}{(7.5\times 10^9)^2} \approx 0.40\ \mathrm{m/s}^2
```

2.2 Disk-UQFF Force Coupling

```
F_{\rm disk} = \rho_{\rm disk} \cdot v_{\rm disk}^2 \cdot (1 + H_0 t) \cdot {\rm SC}_m
```

where:

- ρ_{disk} = protoplanetary disk density at a_{orb} (~10⁻■ kg/m^3 at 11 Myr)
- v_disk = disk gas velocity (~1 km/s at 0.05 AU)
- SC_m = superconductive modifier
- (1 + H■t) = Hubble expansion factor over 11 Myr

For t = 11 Myr = 3.47×10^1■ s:

```
(1 + H_0 t) \approx 1 + 2.27\times 10^{-18} \times 3.47\times 10^{14} \approx 1.0008
```

2.3 UQFF FUBi_i (Planetary Scale)

```
F_{U\_Bi\_i}^{\rm planet} = F_{U\_Bi\_i}(M_{\rm planet}, r_{\rm orbit}, \omega_{\rm act})
```

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3. Key Values

Quantity	Formula	Value
T_age	TESS age estimate	11 Myr
P_orb	TESS + RV	11 days
a_orb	Kepler 3rd law	~0.05 AU
g_tide	GM_star/a^2	~0.40 m/s^2
F_disk	$\rho disk\cdot v^2\cdot SC$m	disk-epoch force
(1+H■t)	Hubble correction	1.0008

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4. Physical Significance

TOI-1227b is exceptional because it is young enough that the UQFF disk-coupling term F*disk is non-negligible: the protoplanetary disk provides a dense medium that couples the SC*m modifier to the local vacuum field. This is impossible for older planetary systems where the disk has dissipated. The UQFF prediction is that disk-embedded planets during the 1–100 Myr epoch receive a systematic F_disk force that contributes at the ~0.1% level to their orbital energy budget, producing a measurable shift in transit timing variations (TTVs) detectable by PLATO/Ariel.

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5. Deduplication Note

- **vs. PAPER_357 vs. Solar System UQFF papers:** All prior UQFF planet papers used mature (>1 Gyr) systems; TOI-1227b at 11 Myr is the youngest planet in the UQFF dataset.
- **Unique:** Disk coupling F*disk* = $\rho disk\cdot v^2\cdot(1+H$■$t)\cdot SC\_m$ is unique to young-planet UQFF.

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6. Classification

Physics Territory: FIRST UQFF young exoplanet (11 Myr) with disk-UQFF SC_m coupling **Scale:** Planetary (sub-Neptune, 0.05 AU orbit) **CP Implementation:** `TOI1227bYoungNeptuneExoplanetFUBiCalculator` (CondensedPhysics4.py, Session 97)
